Hyperbolic Functions: Osborn's Rule & Proof

Questions 1–57

Question 1. By considering Osborn's Rule, or otherwise, write down the analogous hyperbolic identity for each of these trigonometric identities:

a) $\sin^2 x + \cos^2 x \equiv 1$
b) $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
c) $\tan^2 x + 1 \equiv \sec^2 x$
d) $\sin(x + y) \equiv \sin x \cos y + \cos x \sin y$
e) $\sin(x - y) \equiv \sin x \cos y - \cos x \sin y$
f) $\cos(x + y) \equiv \cos x \cos y - \sin x \sin y$
g) $\cos(x - y) \equiv \cos x \cos y + \sin x \sin y$
h) $\tan(x + y) \equiv \dfrac{\tan x + \tan y}{1 - \tan x \tan y}$
i) $\tan(x - y) \equiv \dfrac{\tan x - \tan y}{1 + \tan x \tan y}$
j) $\sin 2x \equiv 2 \sin x \cos x$
k) $\cos 2x \equiv \cos^2 x - \sin^2 x$
l) $\cos 2x \equiv 2 \cos^2 x - 1$
m) $\cos 2x \equiv 1 - 2 \sin^2 x$
n) $\tan 2x \equiv \dfrac{2 \tan x}{1 - \tan^2 x}$
o) $\sin 3x \equiv 3 \sin x - 4 \sin^3 x$
p) $\cos 3x \equiv 4 \cos^3 x - 3 \cos x$
q) $\tan 3x \equiv \dfrac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$
r) $\sin 4x \equiv \cos x(4 \sin x - 8 \sin^3 x)$
s) $\cos 4x \equiv 8 \cos^4 x - 8 \cos^2 x + 1$
t) $\tan 4x \equiv \dfrac{4 \tan x - 4 \tan^3 x}{1 - 6 \tan^2 x + \tan^4 x}$

Question 2. Write the following hyperbolic expressions in terms of $\tanh x$, $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$ only:

a) $\dfrac{\sinh x}{\cosh^2 x}$
b) $\dfrac{\cosh x}{\sinh^2 x}$

Question 3. Simplify the following hyperbolic expressions as far as possible:

a) $\operatorname{cosech} x \sinh x \cosh x$

b) $\tanh x \operatorname{sech} x \cosh x$

c) $\tanh x \operatorname{cosech} x \coth x \sinh x$

Question 4. Simplify the following hyperbolic expressions, removing any quotients where possible:

a) $\dfrac{\operatorname{sech} x}{\operatorname{cosech} x}$

b) $\dfrac{\coth x}{\operatorname{sech} x}$

c) $\dfrac{\operatorname{cosech} x}{\tanh x}$

d) $\dfrac{\operatorname{sech} x}{\coth x}$

e) $\dfrac{\operatorname{sech}^2 x}{\coth x}$

f) $\dfrac{\operatorname{cosech}^2 x \coth^2 x}{\operatorname{sech} x \sinh x}$

Question 5. Given that k is a constant where $k > 1$, show that

$$-\ln(k + \sqrt{k^2 - 1}) = \ln(k - \sqrt{k^2 - 1}).$$

Question 6. Given that k is a constant, show that

$$-\ln(k + \sqrt{k^2 + 1}) \neq \ln(k - \sqrt{k^2 + 1}).$$

Question 7. Show that the equation $\cosh x = k$, where k is a constant such that $k > 1$, has two real solutions: $x = \ln(k + \sqrt{k^2 - 1})$ and $x = \ln(k - \sqrt{k^2 - 1})$.

Question 8. It is given that $\cosh x = 4$. Show that $x = \ln(4 \pm \sqrt{15})$ and hence $x = \pm \ln(4 + \sqrt{15})$.

Question 9. Express each of these expressions in terms of $\cosh x$:

a) $8\cosh^2 x + 7\sinh^2 x$

b) $3\sinh^2 x - 13\cosh^2 x$

Question 10. Solve $3\cosh^2 x - \sinh^2 x = 5$, writing your answers in exact form.

Question 11. Solve each of these equations, writing your answers in exact form:

a) $\cosh^2 x - \sinh x - 7 = 0$
b) $\sinh^2 x - 2\cosh x - 7 = 0$
c) $\cosh^2 x - 14\sinh x + 47 = 0$
d) $2\sinh^2 x - 13\cosh x + 8 = 0$
e) $3\cosh^2 x + 22\sinh x - 19 = 0$
f) $4\sinh^2 x - 4\cosh x - 11 = 0$

Question 12. Solve each of these equations, writing your answers in exact form:

a) $25\cosh^4 x - 40\cosh^2 x - 9 = 0$
b) $36\cosh^4 x - 36\sinh^2 x - 91 = 0$
c) $192\sinh^4 x - 172\cosh^2 x + 67 = 0$

Question 13. Starting from $\cosh^2 x - \sinh^2 x \equiv 1$:

a) derive an identity for $\coth x$ and $\operatorname{cosech} x$

b) derive an identity for $\tanh x$ and $\operatorname{sech} x$

Question 14. Solve each of these equations, writing your answers in exact form:

a) $2\operatorname{sech}^2 x - 3\tanh x = 0$

b) $3\tanh^2 x - 5\operatorname{sech} x - 1 = 0$

c) $\operatorname{cosech}^2 x - 2\coth x - 7 = 0$

d) $4\coth^2 x - 25\operatorname{cosech} x + 2 = 0$

Question 15. Solve each of these equations, writing your answers in exact form:

a) $8\operatorname{sech}^2 x - 6\tanh x + 1 = 0$

b) $10\tanh^2 x - 21\operatorname{sech} x + 3 = 0$

c) $24\operatorname{cosech}^2 x + 58\coth x + 59 = 0$

d) $25\coth^2 x - 73\operatorname{cosech} x = 31$

Question 16. Prove there are no real solutions to the equation $6\tanh^2 x + 17\operatorname{sech} x - 18 = 0$.

Question 17. Solve each of these equations, writing your answers in exact form:

a) $\sinh x = \operatorname{sech} x$

b) $\tanh x = \operatorname{sech} x$

c) $\sinh x = 2\coth x$

d) $3\cosh x = \coth x$

Question 18.

a) Sketch the graphs of $y = 2\cosh x$ and $y = \operatorname{cosech} x$ on the same axes.

b) Determine the exact coordinates of where the two graphs intersect.

Question 19. It is given that $\sinh x = 3$. Find the exact values of:

a) $\cosh x$
b) $\tanh x$
c) $\coth x$
d) $\operatorname{cosech} x$
e) $\operatorname{sech} x$

Question 20. It is given that $\operatorname{sech} x = \dfrac{1}{\sqrt{2}}$. Find the exact values of:

a) $\tanh x$
b) $\cosh x$
c) $\sinh x$
d) $\operatorname{cosech} x$
e) $\coth x$

Question 21. It is given that $48\coth^2 x + 34\operatorname{cosech} x - 53 = 0$ and x is positive. Find the exact values of:

a) $\operatorname{cosech} x$
b) $\sinh x$
c) $\cosh x$
d) $\operatorname{sech} x$
e) $\tanh x$
f) $\coth x$

Question 22. Simplify the following hyperbolic expressions:

a) $\dfrac{\operatorname{sech}^2 x}{1 - \tanh^2 x}$
b) $\dfrac{\coth^2 x - 1}{\operatorname{cosech}^2 x}$
c) $\dfrac{1 - \tanh^2 x}{\coth^2 x - 1}$
d) $\dfrac{3 - 3\coth^2 x}{8 - 8\tanh^2 x}$

Question 23. It is given that $\sinh(1+\ln 6) = \dfrac{36e^2-1}{12e}$.

a) Hence find the exact value of $\cosh(1+\ln 6)$ in terms of e.

b) Using your answer to (a), show that $\tanh(1+\sqrt{6}) = \dfrac{36e^2-1}{36e^2+1}$.

Question 24. It is given that $p(x) = 12x^3 - 7x^2 - 42x + 40$.

a) Show that $(4x-5)$ is a factor of $p(x)$.

b) Write $p(x)$ as a product of three linear factors.

c) Hence prove that there are no real solutions to the equation

$$12\operatorname{sech}^3\theta + 7\tanh^2\theta - 42\operatorname{sech}\theta + 33 = 0.$$

Question 25.

a) Find $\sinh(\ln 3)$ and $\cosh(\ln 3)$.

b) Find $\sinh(\ln 4)$ and $\cosh(\ln 4)$.

c) Using your answers to (a) and (b), find the exact value of $\sinh(\ln 4 - \ln 3)$.

Question 26.

a) Find $\sinh(\ln 5)$ and $\cosh(\ln 5)$.

b) Find $\sinh(\ln 6)$ and $\cosh(\ln 6)$.

c) Using your answers to (a) and (b), find the exact value of $\cosh(\ln 6 - \ln 5)$.

Question 27.

a) Find $\tanh(\ln 2)$.

b) Find $\tanh(\ln 7)$.

c) Using your answers to (a) and (b), find the exact value of $\tanh(\ln 2 + \ln 7)$.

Question 28. Write the following in the form $\sinh k$, where k is a constant:

a) $\sinh 3 \cosh 6 - \cosh 3 \sinh 6$

b) $\sinh 4 \cosh 5 + \cosh 4 \sinh 5$

Question 29. Write the following in the form $\cosh k$, where k is a constant:

a) $\cosh 3 \cosh 7 - \sinh 3 \sinh 7$

b) $\cosh 8 \cosh 9 + \sinh 8 \sinh 9$

Question 30. It is given that $\cosh x = 2$ and $\cosh y = 3$, where x and y are real. Find the exact value of:

a) $\cosh(x - y)$

b) $\sinh(x - y)$

Question 31. Simplify the following expressions:

a) $\sinh(x + \ln 2) + \cosh(x - \ln 2)$

b) $\sinh(x - \ln 3) - \cosh(x + \ln 3)$

c) $\cosh(x - \ln 4) - \sinh(x - \ln 4)$

Question 32. In each case, find the exact value that solves the equation:

a) $\sinh(x + \ln 2) = \cosh x$

b) $\cosh(x - \ln 3) = \cosh x$

c) $\cosh(x - \ln 4) = \sinh x$

Question 33. In each case, find the exact value that solves the equation:

a) $\sinh(x + \ln 2) = \cosh(x - \ln 4)$

b) $\cosh(x - \ln 9) = \sinh(x + \ln 3)$

c) $\cosh(x + 3\ln 2) = \cosh(x - \ln 20)$

Question 34. Prove the following hyperbolic identities:

a) $\coth x + \cosh x \equiv \coth x(1 + \sinh x)$

b) $\coth x + \operatorname{sech} x \equiv \operatorname{cosech} x \operatorname{sech} x(\sinh x + \cosh^2 x)$

c) $\coth x - \tanh x \equiv \operatorname{cosech} x \operatorname{sech} x$

Question 35. Prove the following hyperbolic identities:

a) $\cosh x - \operatorname{sech} x \equiv \sinh x \tanh x$

b) $\operatorname{cosech} x + \sinh x \equiv \cosh x \coth x$

c) $\operatorname{cosech} x \operatorname{sech} x \equiv \tanh x(\coth^2 x - 1)$

d) $\dfrac{\operatorname{cosech} x(\sinh x \tanh x - 1)}{\sinh x + \cosh x} \equiv \tanh x - \operatorname{cosech} x$

Question 36. Simplify the following expressions:

a) $8 \sinh 2x \cosh 2x$

b) $2 \cosh^2 5x - 1$

c) $1 + 2 \sinh^2 3x$

d) $9 \cosh^2 \frac{x}{4} + 9 \sinh^2 \frac{x}{4}$

Question 37. Solve the following equations, giving your answers in an exact form:

a) $\sinh 2\theta - 2\sinh\theta = 0$

b) $\sinh 2\theta = 3\sinh\theta$

c) $5\sinh 2\theta = 6\cosh\theta$

d) $8\cosh\theta - 3\sinh 2\theta = 0$

Question 38. Solve the following equations, giving your answers in an exact form:

a) $2\cosh 2\theta = 7\cosh\theta$

b) $3\cosh 2\theta + 11\sinh\theta = 0$

c) $3\sinh\theta - \cosh 2\theta = 0$

d) $\cosh 2\theta - 4\cosh\theta = 0$

Question 39. Solve the following equations, giving your answers in an exact form:

a) $\cosh 2\theta + 2\sinh\theta - 5 = 0$

b) $5\cosh 2\theta - 3\sinh\theta = 5$

c) $8\cosh\theta - 3\cosh 2\theta = 3$

d) $5\cosh\theta - 12\cosh 2\theta + 17 = 0$

Question 40. Solve the following equations, giving your answers in an exact form:

a) $2\sinh 2\theta\,\operatorname{cosech}\theta = 5$

b) $5\sinh 2\theta\,\operatorname{sech}\theta = 7$

c) $\cosh 2\theta\,\operatorname{cosech}\theta = 3$

d) $2\cosh 2\theta\,\operatorname{sech}\theta = 7$

Question 41. Solve the following equations, giving your answers in an exact form:

a) $5\tanh 2x - 9\tanh x = 0$

b) $4\tanh x - 3\tanh 2x = 0$

Question 42. It is given that $\sinh x = \frac{4}{3}$. Find the exact value of $\sinh 2x$.

Question 43. It is given that $\sinh x = -\frac{1}{3}$. Find the exact value of $\cosh 2x$.

Question 44. It is given that $\cosh x = \frac{9}{4}$. Find the exact value of $\tanh 2x$.

Question 45.

a) Use hyperbolic identities, including $\sinh(A \pm B) \equiv \sinh A \cosh B \pm \cosh A \sinh B$, to write $\sinh 3\theta$ in terms of $\sinh \theta$.

It is given that $x = \sinh \theta$ and that $20x^3 + 15x - 12 = 0$.

b) Find the value of θ in the form $p \ln q$ where p is a rational number and q is a positive integer.

c) Hence find the real root of the equation $20x^3 + 15x - 12 = 0$, giving your answer in the form $\frac{1}{2}(5^m - 5^n)$, where m and n are rational numbers.

Question 46. Prove the following hyperbolic identities:

a) $\coth^2 x - \tanh^2 x \equiv \operatorname{cosech}^2 x + \operatorname{sech}^2 x$

b) $\sinh^2 x \tanh^2 x \equiv \sinh^2 x - \tanh^2 x$

c) $\operatorname{cosech}^2 x - \operatorname{sech}^2 x \equiv 4 \operatorname{cosech}^2 2x$

Question 47. Prove the following hyperbolic identities:

a) $(\operatorname{sech} x - \tanh x)^2 \equiv 1 - 2\operatorname{sech} x \tanh x$

b) $(\operatorname{sech} x - \coth x)^2 \equiv 2 - \tanh^2 x - 2\operatorname{cosech} x + \operatorname{cosech}^2 x$

c) $(\operatorname{sech} x - \operatorname{cosech} x)^2 \equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x(\cosh 2x - \sinh 2x)$

Question 48.

a) Prove the following hyperbolic identity:

$$\frac{1+\cosh x}{\sinh x} - \frac{\sinh x}{1+\cosh x} \equiv 2\operatorname{cosech} x.$$

b) Hence, where $t \neq 0$, solve the equation

$$\frac{1+\cosh t}{\sinh t} - \frac{\sinh t}{1+\cosh t} = -4.$$

Question 49.

a) Prove the following hyperbolic identity:

$$\frac{\sinh x}{\cosh x - 1} - \operatorname{cosech} x \equiv \coth x.$$

b) Hence, given $t \neq 0$, solve the equation

$$\frac{\sinh t}{\cosh t - 1} = 8 + \coth t.$$

Question 50. Show that the solution to the equation $\operatorname{sech} \frac{x}{2} = 6 \sinh \frac{x}{2}$ can be written in the form $\ln(p + \sqrt{q}) - \ln r$, where p, q and r are positive integers.

Question 51. Prove the following hyperbolic identities:

a) $\coth 2x - \tanh 2x \equiv 2\,\text{cosech}\, 4x$

b) $\coth x - \tanh 2x \equiv \coth x\,\text{sech}\, 2x$

Question 52. Prove the following hyperbolic identities:

a) $\coth 2x - \cosh 2x \equiv \coth 2x(1 - \sinh 2x)$

b) $\coth 2x + \text{cosech}\, 2x \equiv \coth x$

Question 53. Prove the following hyperbolic identity:

$$2\operatorname{cosech} 4x(\sinh^2 x + \cosh 2x) \equiv \tanh 2x + \operatorname{cosech} 2x.$$

Question 54. Prove the following hyperbolic identities:

a) $(\sinh 2x + \cosh 2x)^2 \equiv \sinh 4x + \cosh 4x$

b) $\dfrac{2\sinh x}{\cosh 2x - 1} - \dfrac{2\sinh x}{\cosh 2x + 1} \equiv \operatorname{cosech} x \operatorname{sech}^2 x$

Question 55. Prove the following hyperbolic identities:

a) $\cosh 2x - \sinh 2x \equiv \dfrac{1}{(\sinh x + \cosh x)^2}$

b) $\coth x(\cosh 2x - \sinh 2x) \equiv \dfrac{\cosh x(\coth x - 1)}{\sinh x + \cosh x}$

Question 56. Solve the equation $8\sinh x + 15\cosh x = 6\sinh x \cosh x + 20$, giving your answer in exact form.

Question 57. Solve the following equations:

a) $\sinh(\ln(8+2x)) = x$

b) $\cosh(\ln(3-5x)) + 10x = 0$

c) $\operatorname{sech}(\ln(4x+9)) + \frac{1}{12x} = 0$

d) $\tanh(\ln(\sqrt{3}+4x)) = 2x$